College Club Management System (CCMS)

Classified Software Test Specification & Execution Report

Course: 25CS312 (Software Engineering) **Date:** November 2026

Frameworks: Python 3.14 • Pytest 9.1.1 • Selenium 4 **Result: 40 / 40 Passed (100% Pass Rate)**

Testing Category	Sub-Division Directory	Test Focus & Scope	Total Cases	Status
Black Box Testing	TESTING/black_box/	Positive, Negative, Boundary, Functional workflows	**22**	**22 / 22 PASS**
White Box Testing	TESTING/white_box/	Database integrity, Calculation branches, Role matrix	**9**	**9 / 9 PASS**
Selenium 4 Testing	TESTING/selenium/	Headless Chrome E2E browser UI, Theme toggle, Forms	**9**	**9 / 9 PASS**
Total Classified Suite	**Full System Verification**	**End-to-End Multi-Tier Software Quality**	**40**	**40 / 40 PASS (100%)**

1. Black Box Testing Suite (22 Test Cases)

TC ID	Module / Method	Test Description & Objective	Expected Result	Status
TC-BB-POS-01	test_bb_pos_01_landing_page_accessible	Verify landing page loads with HTTP 200 and CCMS title	HTTP 200 OK, title present	**PASS**
TC-BB-POS-02	test_bb_pos_02_clubs_catalog_route_accessible	Verify /clubs catalog route is publicly reachable	HTTP 200 OK	**PASS**
TC-BB-POS-03	test_bb_pos_03_auth_login_route_accessible	Verify authentication /login route is available	HTTP 200 OK	**PASS**
TC-BB-POS-04	test_bb_pos_04_auth_register_route_accessible	Verify registration /register route is available	HTTP 200 OK	**PASS**
TC-BB-POS-05	test_bb_pos_05_supabase_categories_read	Verify reading categories returns active category list	HTTP 200, non-empty list	**PASS**
TC-BB-POS-06	test_bb_pos_06_supabase_clubs_read	Verify reading approved clubs catalog	HTTP 200, approved clubs	**PASS**
TC-BB-POS-07	test_bb_pos_07_frontend_static_assets_serve	Verify serving HTML5 doctype and ES module tags	HTML5 valid syntax	**PASS**
TC-BB-NEG-01	test_bb_neg_01_login_invalid_password	Verify wrong password authentication rejection	HTTP 400 Bad Request	**PASS**
TC-BB-NEG-02	test_bb_neg_02_login_nonexistent_user	Verify unregistered email login failure	HTTP 400/422 rejection	**PASS**
TC-BB-NEG-03	test_bb_neg_03_unauthorized_profile_role_update_blocked	Verify anonymous user cannot update roles	HTTP 401/403 or empty	**PASS**
TC-BB-NEG-04	test_bb_neg_04_direct_insert_attendance_unauthorized	Verify anonymous insert into attendance blocked	HTTP 400/401/403 blocked	**PASS**
TC-BB-NEG-05	test_bb_neg_05_nonexistent_club_slug_query	Verify non-existent club slug lookup handling	HTTP 200 with [] empty set	**PASS**
TC-BB-BND-01	test_bb_bnd_01_event_capacity_zero_boundary	Verify capacity = 0 marks event closed	Capacity = 0 observed	**PASS**
TC-BB-BND-02	test_bb_bnd_02_event_capacity_exact_limit_boundary	Verify capacity ceiling boundary (count >= max)	is_full returns True	**PASS**
TC-BB-BND-03	test_bb_bnd_03_search_empty_string_boundary	Verify empty search term returns all active clubs	HTTP 200 full list	**PASS**
TC-BB-BND-04	test_bb_bnd_04_search_extreme_length_string_boundary	Verify 2,500-char extreme query buffer safety	No server crash (200/400)	**PASS**
TC-BB-BND-05	test_bb_bnd_05_malformed_uuid_boundary	Verify boundary malformed UUID input handling	HTTP 400 validation error	**PASS**
TC-BB-FNC-01	test_bb_fnc_01_membership_lifecycle_valid_states	Verify membership lifecycle state domain	States in allowed set	**PASS**
TC-BB-FNC-02	test_bb_fnc_02_attendance_status_domain	Verify attendance domain (present, absent, late)	Domain verified	**PASS**
TC-BB-FNC-03	test_bb_fnc_03_spa_client_fallback_handling	Verify deep link routing serves index.html	index.html served	**PASS**
TC-BB-FNC-04	test_bb_fnc_04_empty_auth_payload_rejection	Verify empty JSON auth payload rejected	HTTP 400 rejection	**PASS**
TC-BB-FNC-05	test_bb_fnc_05_malformed_json_recovery	Verify recovery on non-JSON raw strings	HTTP 400/415 handled	**PASS**

2. White Box Testing Suite (9 Test Cases)

TC ID	Module / Method	Test Description & Objective	Expected Result	Status
TC-WB-DB-01	test_wb_db_01_foreign_key_constraint_cascade	Verify relational FK constraints block orphan categories	Cascade constraint error	**PASS**
TC-WB-DB-02	test_wb_db_02_postgrest_cors_header_contracts	Verify server CORS preflight contracts and origin	Access-Control headers	**PASS**
TC-WB-DB-03	test_wb_db_03_auth_service_health_endpoint	Inspect internal Supabase GoTrue Auth service status	Auth service responsive	**PASS**
TC-WB-DB-04	test_wb_db_04_frontend_backend_connectivity	Verify presentation and data tier connectivity	HTTP 200 bidirectional	**PASS**
TC-WB-CALC-01	test_wb_calc_01_attendance_rate_branch_coverage	Full branch coverage of attendance percentage formula	0%, 100%, 66.7% math	**PASS**
TC-WB-CALC-02	test_wb_calc_02_department_sorting_logic	Verify internal array grouping by student department	Department arrays accurate	**PASS**
TC-WB-CALC-03	test_wb_calc_03_password_length_constraint_eval	Verify internal 6-character password constraint branch	Boolean bounds correct	**PASS**
TC-WB-ROLE-01	test_wb_role_01_strict_dual_role_matrix	Validate student vs coordinator permission matrix	Role matrix strictness	**PASS**
TC-WB-ROLE-02	test_wb_role_02_club_taxonomy_structural_integrity	Verify database taxonomy has no orphan clubs	All club categories valid	**PASS**

3. Selenium 4 End-to-End Browser Testing Suite (9 Test Cases)

TC ID	Module / Method	Test Description & Objective	Expected Result	Status
TC-SEL-01	test_sel_01_landing_page_title_and_branding	Headless Chrome page render, title & brand marker	Title contains 'CCMS'	**PASS**
TC-SEL-02	test_sel_02_theme_toggle_interaction	Interactive theme toggle click (Light <-> Dark mode)	DOM/title attribute toggles	**PASS**
TC-SEL-03	test_sel_03_navigation_links_presence	Verify top navigation bar links are displayed	Portals, Timetable, Clubs	**PASS**
TC-SEL-04	test_sel_04_clubs_catalog_navigation_and_cards	Browser navigation to /clubs and club card render	Club cards rendered	**PASS**
TC-SEL-05	test_sel_05_category_filter_pills_interactive	Verify domain filter pill elements and search input	Inputs/buttons displayed	**PASS**
TC-SEL-06	test_sel_06_login_page_form_elements	Inspect /login email, password, and submit button	Form controls displayed	**PASS**
TC-SEL-07	test_sel_07_register_page_form_elements_and_role_selection	Inspect /register form fields and role selector	Role tabs rendered	**PASS**
TC-SEL-08	test_sel_08_activity_timetable_cards_display	Verify weekly timetable section and schedule cards	Monday/Wed/Sat slots	**PASS**
TC-SEL-09	test_sel_09_dual_portal_access_cards	Verify Student & Coordinator portal cards display	Both portal cards visible	**PASS**